

## Introduction to Cyber Security and Basic Terminologies

Cyberspace is the virtual, interconnected environment of digital networks and devices. Cybercrime involves illegal acts committed using these systems, often for financial gain, while Cyber Security consists of technologies and practices designed to protect this digital infrastructure from attacks, ensuring confidentiality, integrity, and availability.

### 1. Cyberspace

- A global, intangible domain comprised of the internet, computer systems, networks, and data, allowing for digital communication and interaction.
- Cyberspace is termed as a virtual and dynamic domain created by computer clones. Cyberspace best describes the immaterial space where interactions through digital networks, the internet, and computer systems take place

### 2. Cybercrime

- Criminal activities where a computer or network is either the tool, target, or place of the crime.
- Cyber Crime relates to the word cyber which encompasses the computer network, using which one can perform any activity in the real time world. Cyber Crimes such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy use digital data from Computer systems and other electronic devices. Cybercrime refers to the use of a computer to commit a crime. As a criminal activity, it began when hackers started illegally accessing high-level computer networks.
- Cybercriminals are individuals or teams of people who use computer, devices and/or malicious software to gain unauthorized access to hardware, operating systems or networks. Their aim is to disrupt, damage, influence, steal data, steal or extort money.

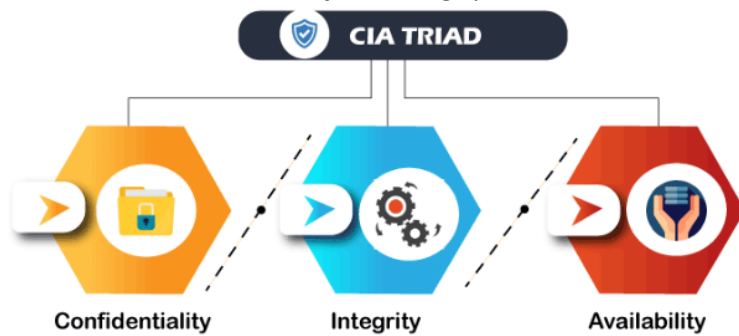
Types: Identity Thieves, Internet Stalkers, Phishing Scammers, Cyber Terrorists

- **Types:**
  - Hacking: Unauthorized access to systems. System hacking means using computers to commit fraudulent acts such as fraud, privacy invasion, stealing corporate/personal data, etc. Cybercrimes cost many organizations millions of dollars every year. Businesses need to protect themselves against such attacks. Hacking refers to the misuse of devices like computers, smartphones, tablets, and networks to cause damage to or corrupt systems, gather information on users, steal data and documents, or disrupt data-related activity.
  - Phishing/Social Engineering: Deceiving users to steal credentials.
  - Malware/Ransomware: Malicious software designed to damage or lock data.
  - Identity Theft/Financial Fraud: Stealing personal information for monetary gain. Identity theft occurs when someone uses another person's personal identifying information, like their name, identifying number, or credit card number, without their permission, to commit fraud or other crimes.
  - Stalking: It is the use of the Internet or other electronic means to stalk or harass an individual, group, or organization. It may include false accusations, defamation, slander and libel. It may also include monitoring, identity theft, threats, vandalism, and solicitation for sex, or blackmail. Types -

### 3. Cyber Security

- The protection of internet-connected systems—including hardware, software, and data—from cyberattacks.
- Cyber security is the practice of protecting systems, networks, and programs from digital attacks. These cyber-attacks are usually aimed at accessing, changing, or destroying sensitive information; extorting money from users; or interrupting normal business processes.
- It is the application of technologies, processes and controls to protect systems, networks, programs, devices and data from cyber-attacks.
- **Core Principles (CIA Triad):**
- Cyber Security's main objective is to ensure data protection. The security community provides a triangle of three related principles to protect the data from cyber-attacks. This principle is called the CIA triad.
- The CIA model is designed to guide policies for an organization's information security infrastructure. When any security breaches are found, one or more of these principles has been violated.

- **Confidentiality:** Ensuring only authorized users access data.
- **Integrity:** Maintaining the accuracy and reliability of data.
- **Availability:** Ensuring systems are accessible when needed.

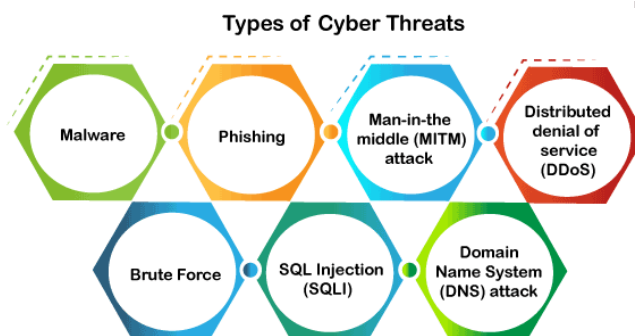


### Cyber Security Attack Steps:

1. Reconnaissance (information gathering)
2. Type of attack – active/ passive
3. Scanning and scrutinizing the gathered information for the validity of the information as well as to identify the existing vulnerabilities.
4. Launching an attack (gaining and maintaining the system access).

### Types of Cyber Security Threats

A threat in cyber security is a malicious activity by an individual or organization to corrupt or steal data, gain access to a network, or disrupts digital life in general. The cyber community defines the following threats available today:



### Malware:

Malware is malicious software and refers to any software that is designed to cause harm to computer systems, networks, or users. Malware can take many forms.

It includes a variety of harmful software intended to infiltrate and compromise systems. Understanding malware's complexities is essential for individuals and businesses to respond and prevent these malicious activities effectively.

### Types:

**a. Spyware:** Spyware is any software that installs itself on your computer and starts covertly monitoring your online behavior without your knowledge or permission. For example, by violating their privacy or endangering their device's security. Its purpose is to steal private information from a computer system for a third party. Spyware collects information and sends it to the hacker that continuously "spies" on users and their activities.

### b. Virus & Worm:

**Virus:** - A computer virus is a type of computer program that, when executed, replicates itself by modifying other computer programs and inserting its own code. If this replication succeeds, the affected areas are then said to be "infected" with a computer virus, a metaphor derived from biological viruses.

- Computer viruses generally require a host program. The virus writes its own code into the host program. When the program runs, the written virus program is executed first, causing infection and damage.

**Worm:** A computer worm is a standalone malware computer program that replicates itself in order to spread to other computers. It often uses a computer network to spread itself, relying on security failures on the target computer to access it.

**c. Trojan:** It is a malware that carries out malicious operations under the appearance of a desired operation such as playing an online game. Trojan horse programs are malware that is disguised as legitimate software. A Trojan horse program will hide on your computer until it's called upon.

**d. Ransomware:** Ransomware grasps a computer system or the data it contains until the victim makes a payment. Ransomware encrypts data in the computer with a key that is unknown to the user. The user has to pay a ransom (price) to the criminals to retrieve data. Once the amount is paid the victim can resume using his/her system.

**e. Adware:** It displays unwanted ads and pop-ups on the computer. It comes along with software downloads and packages. It generates revenue for the software distributor by displaying ads.

**Phishing:** Phishing is a type of attack where an attacker sends a fraudulent ("spoofed") message designed to trick a human victim into revealing sensitive information to the attacker or to deploy malicious software on the victim's infrastructure like ransomware. Phishing attacks have become increasingly sophisticated and often transparently mirror the site being targeted, allowing the attacker to observe everything while the victim is navigating the site, and transverse any additional security boundaries with the victim.

#### TYPES OF PHISHING



**Denial of Service (DoS) Attack:** Denial-of-service attack (DoS attack) is a cyber-attack in which the perpetrator seeks to make a machine or network resource unavailable to its intended users by temporarily or indefinitely disrupting services of a host, connected to the Internet. Denial of service is typically accomplished by flooding the targeted machine or resource with excessive requests in an attempt to overload systems and prevent some or all legitimate requests from being fulfilled.

**Distributed denial-of-service attack (DDoS):** In a distributed denial-of-service attack (DDoS attack), the incoming traffic flooding the victim originates from many different sources. This effectively makes it impossible to stop the attack simply by blocking a single source. A DDoS attack is a distributed denial-of-service attack wherein the victim's computer is bombarded with traffic from multiple systems across different locations. The focus is to burden the victim's server such that it stops working as expected. If it is a network, the server becomes inaccessible to all its users.

**SQL Injection:** It is a code injection technique used to attack data-driven applications, in which malicious SQL statements are inserted into an entry field for execution (e.g. to dump the database contents to the attacker). SQL injection is a web security vulnerability that allows an attacker to interfere with the queries that an application makes to its database. It generally allows an attacker to view data that they are not normally able to retrieve.

**Man-in-the-middle (MitM) attack:** It is a form of cyberattack in which criminals exploiting weak web-based protocols, insert themselves between entities in a communication channel to steal data.

None of the parties sending email, texting, or chatting on a video call are aware that an attacker has inserted their presence into the conversation and that the attacker is stealing their data.

| Information Security                                                                                                                                                                                       | Cybersecurity                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Information security is a process of protecting information against unauthorized access that could result in information loss or undesired modification.</li> </ul> | <ul style="list-style-type: none"> <li>Cybersecurity refers to the practice of protecting data, its related technologies, and storage sources from cyber threats.</li> </ul>                  |
| <ul style="list-style-type: none"> <li>It primarily focuses on information's confidentiality, integrity, and availability.</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>Primarily focuses on defending data, network, required resources from cyber attacks.</li> </ul>                                                        |
| <ul style="list-style-type: none"> <li>Information security deals with the protection of both forms of information - digital and analog - regardless of the realm.</li> </ul>                              | <ul style="list-style-type: none"> <li>Cybersecurity deals with protecting anything and everything present in the cyber realm including data, devices and associated technologies.</li> </ul> |
| <ul style="list-style-type: none"> <li>Professionals involved with information security forms the foundation of data security from any type of threat.</li> </ul>                                          | <ul style="list-style-type: none"> <li>Cybersecurity professionals deal specifically persistent threats of cyber attacks.</li> </ul>                                                          |
| <ul style="list-style-type: none"> <li>Information security deals with all forms of threats against information only.</li> </ul>                                                                           | <ul style="list-style-type: none"> <li>Cybersecurity on the contrary deals with dangers lurking in cyberspace.</li> </ul>                                                                     |
| <ul style="list-style-type: none"> <li>Disclosure modification and disruption, and unauthorized access are the most important issues tackled by information security.</li> </ul>                           | <ul style="list-style-type: none"> <li>Cybersecurity deals with cybercrimes, cyber frauds, and law enforcement in cyberspace.</li> </ul>                                                      |